

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL A-LEVEL CHEMISTRY (9620)

Unit 3: Inorganic 2 and Physical 2

Tuesday 14 January 2025 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

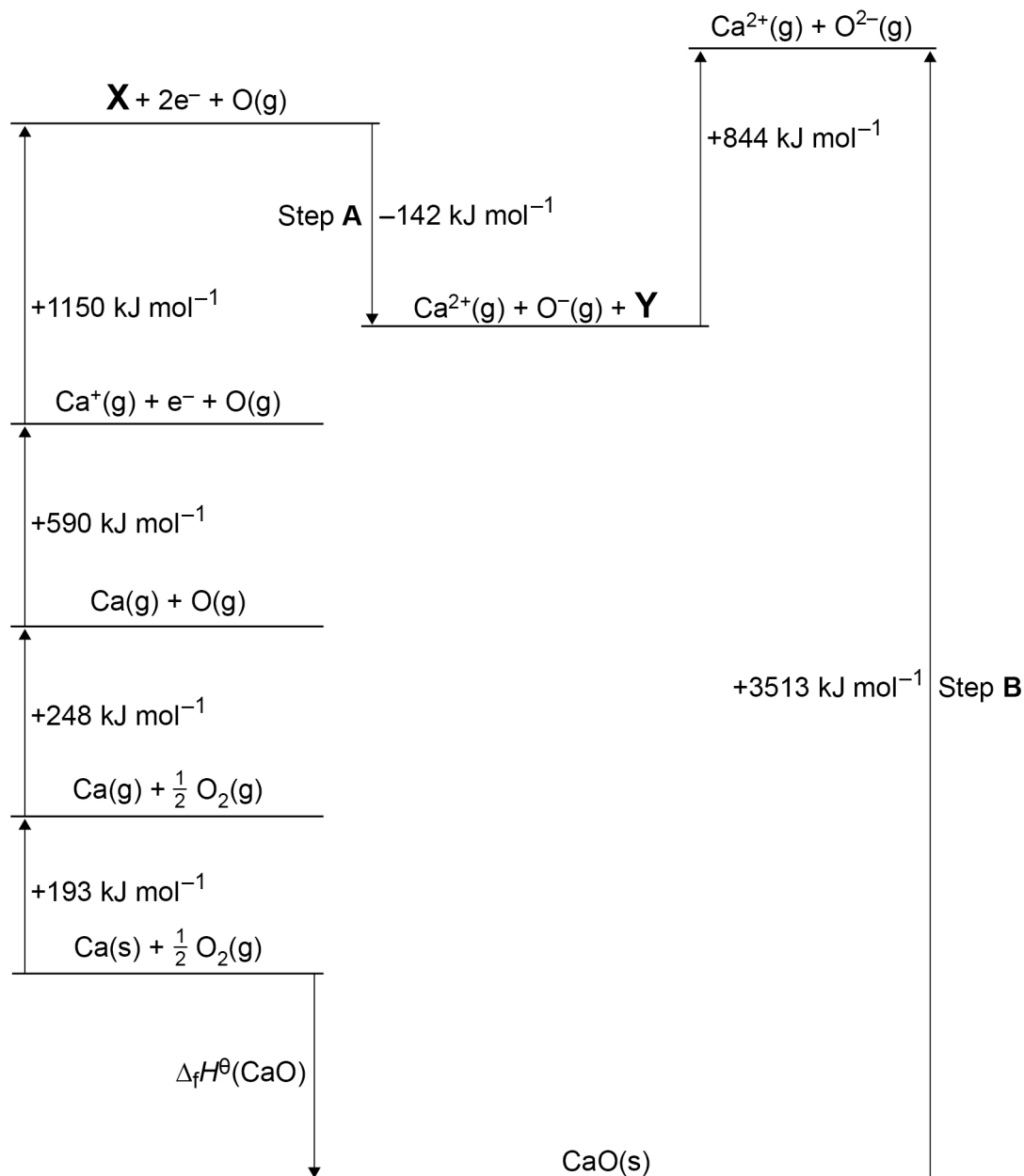
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

Figure 1 shows the Born–Haber cycle for calcium oxide.**Figure 1**

0 1 . 1

Identify species **X** and species **Y**.
Include state symbols where appropriate.**[1 mark]**Species **X** _____Species **Y** _____

0 1 . 2

Give the names of the enthalpy changes represented by Step **A** and Step **B**.**[2 marks]**Step **A** _____Step **B** _____

0 1 . 3

Use **Figure 1** to calculate the standard enthalpy of formation of calcium oxide.**[2 marks]** $\Delta_f H^\ominus(\text{CaO})$ _____ kJ mol^{-1}

5**Turn over for the next question****Turn over ►**

0	2
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A student does an experiment to determine the enthalpy of solution of magnesium chloride.

Method

- 25.0 cm³ of distilled water are transferred into a plastic cup.
- 1.57 g of magnesium chloride are added to the water in the cup.
- The mixture is stirred so that the magnesium chloride dissolves.
- The temperature of the mixture is monitored using a thermometer.

The temperature rise is 14.2 °C

0	2	.	1
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Calculate a value for the enthalpy of solution, in kJ mol⁻¹, of magnesium chloride.

The specific heat capacity of water = 4.18 J K⁻¹ g⁻¹

Assume that the density of the solution = 1.00 g cm⁻³

[3 marks]

Enthalpy of solution _____ kJ mol⁻¹

0	2	.	2
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Explain why the enthalpy of solution of magnesium hydroxide cannot be determined using the same method.

[1 mark]



Table 1 shows some enthalpy change data.

Table 1

	$\Delta H / \text{kJ mol}^{-1}$
Enthalpy of hydration of magnesium ions	−1920
Enthalpy of hydration of chloride ions	−364
Lattice dissociation enthalpy of magnesium chloride	+2493

0 2 . 3

Use the data in **Table 1** to calculate the enthalpy of solution, in kJ mol^{-1} , of magnesium chloride.

[2 marks]

Enthalpy of solution _____ kJ mol^{-1}

6

Turn over ►



0 3

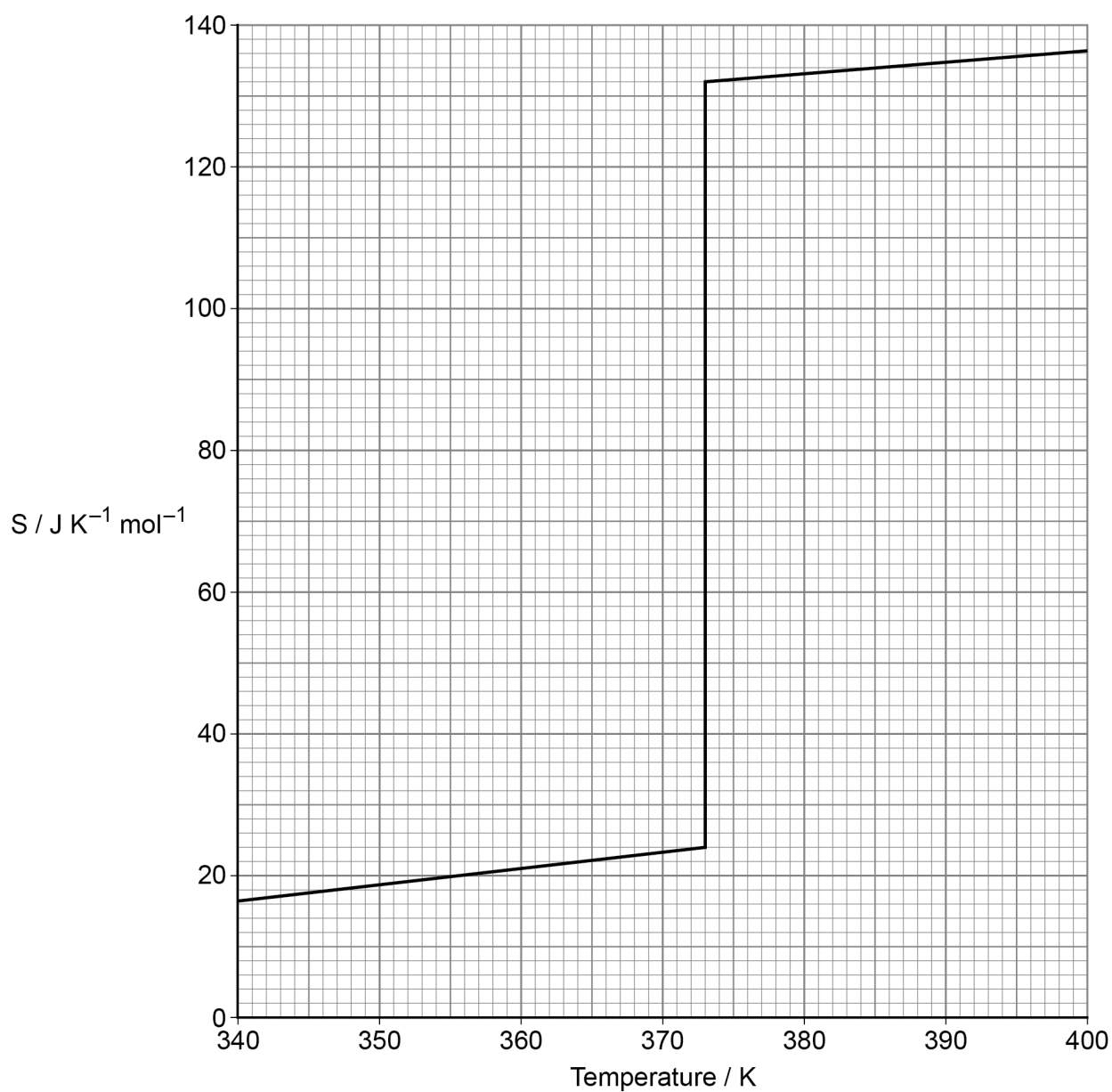
This question is about water.

0 3 . 1

State why the entropy of water is zero at a temperature of 0 K

[1 mark]

Figure 2 shows how the entropy of a sample of water varies with temperature.

Figure 2

03.2 Use **Figure 2** to determine the entropy change, in $\text{J K}^{-1} \text{mol}^{-1}$, when water boils.

[1 mark]

Entropy change _____ $\text{J K}^{-1} \text{mol}^{-1}$

03.3 When a substance boils, the free-energy change, ΔG , is equal to zero.

Use your answer to Question **03.2** to determine the enthalpy change, in kJ mol^{-1} , when water boils.

(If you were unable to answer Question **03.2**, use entropy change = $65.0 \text{ J K}^{-1} \text{mol}^{-1}$
This is **not** the correct value.)

[2 marks]

Enthalpy change _____ kJ mol^{-1}

03.4 **Figure 2** shows the entropy change for water at 373 K
There is also an entropy change at 273 K

State why the entropy change at 373 K is much larger than the
entropy change at 273 K

[1 mark]

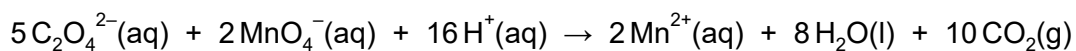
5

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0 4

$\text{Na}_2\text{C}_2\text{O}_4$ can be used to determine the concentration of a solution of KMnO_4



Method

- 0.910 g of anhydrous $\text{Na}_2\text{C}_2\text{O}_4$ is dissolved in some distilled water and the solution is made up to 150 cm^3 using dilute sulfuric acid in a volumetric flask.
- 25.0 cm^3 of this solution are transferred to a conical flask and the solution is warmed to $60\text{ }^{\circ}\text{C}$
- KMnO_4 solution is added from a burette until there is a permanent colour change.

22.35 cm^3 of KMnO_4 solution are needed.

0 4

1

Determine the concentration, in mol dm^{-3} , of the $\text{Na}_2\text{C}_2\text{O}_4$ solution.
Calculate the concentration, in mol dm^{-3} , of the KMnO_4 solution.

[5 marks]

Concentration of $\text{Na}_2\text{C}_2\text{O}_4$ _____ mol dm^{-3}

Concentration of KMnO_4 _____ mol dm^{-3}



0 4 . 2 State the colour change at the end point.

[1 mark]

0 4 . 3 The solution in the conical flask is warmed to 60 °C before the KMnO_4 is added because the reaction is slow.

State why this reaction is slow.

[1 mark]

This reaction is an example of an autocatalysed reaction.

0 4 . 4 Write two equations to show how the reaction between $\text{C}_2\text{O}_4^{2-}$ ions and MnO_4^- ions is catalysed.

[2 marks]

Equation 1

Equation 2

9

Turn over for the next question

Turn over ►



0	5
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This question is about acids, bases and pH

0	5	.	1
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Calculate the pH of the solution formed when 15.0 cm^3 of distilled water are added to 25.0 cm^3 of $0.150 \text{ mol dm}^{-3}$ hydrochloric acid.
Give your answer to **two** decimal places.

[3 marks]

pH _____

0	5	.	2
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Calculate the pH of a $0.650 \text{ mol dm}^{-3}$ solution of barium hydroxide at 298 K
Give your answer to **two** decimal places.

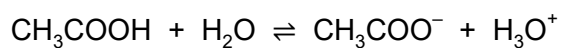
The ionic product of water, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 298 K

[3 marks]

pH _____



Ethanoic acid is a weak acid and ionises in water as shown.



For ethanoic acid, the acid dissociation constant, $K_a = 1.76 \times 10^{-5} \text{ mol dm}^{-3}$ at 298 K

0 5 . 3

What is the effect on the value of this K_a if the concentration of a solution of ethanoic acid is increased?

Tick (✓) **one** box.

[1 mark]

Increases

☐

Remains constant

☐

Decreases

☐

0 5 . 4

Calculate the mass, in g, of sodium ethanoate (CH_3COONa) that needs to be dissolved in 25.0 cm^3 of $0.150 \text{ mol dm}^{-3}$ ethanoic acid to produce a solution with $\text{pH} = 4.50$ at 298 K

[4 marks]

Mass of sodium ethanoate _____ g

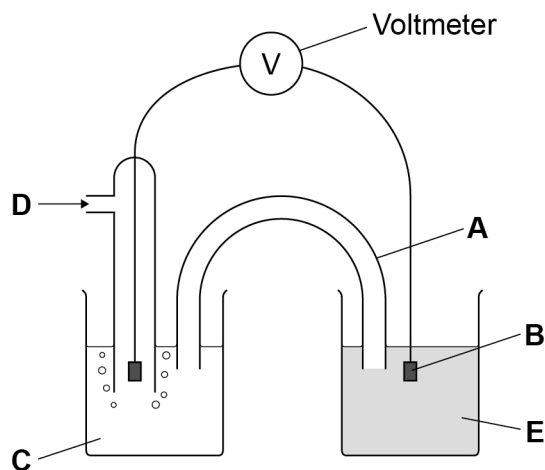
11

Turn over ►



0 6

Figure 3 represents the cell used to measure the standard electrode potential for the $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) / \text{Cr}^{3+}(\text{aq})$ electrode.

Figure 3

0 6 . 1

Name the piece of equipment, **A**.

State the purpose of **A**.

[2 marks]

Name of **A** _____

Purpose of **A** _____

0 6 . 2

Name the substance used as **B**.

[1 mark]

0 6 . 3

The cell is operated at 298 K

Complete **Table 2** to identify substances **C** and **D**.

Include **one** essential condition for each substance.

[2 marks]**Table 2**

	Identity	Condition
C		
D		



0 6 . 4

A student suggested that solution **E** should be prepared by mixing equal volumes of 1.00 mol dm^{-3} aqueous acidified potassium dichromate(VI) solution and 1.00 mol dm^{-3} aqueous chromium(III) sulfate solution.

Describe how the mixture should be changed to allow the standard electrode potential for the $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) / \text{Cr}^{3+}(\text{aq})$ electrode to be measured.

Justify your answer.

[2 marks]

How mixture should be changed _____

Justification _____

The standard electrode potential, $E^\ominus$, for the $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) / \text{Cr}^{3+}(\text{aq})$ electrode is +1.33 V

0 6 . 5

Give the conventional representation for this cell.

[2 marks]

0 6 . 6

Give the ionic equation that shows the overall reaction for this cell.

[1 mark]

0 6 . 7

The electrode on the left in **Figure 3** is replaced by a $\text{Cl}_2 / \text{Cl}^-$ electrode. The reading on the voltmeter changes to -0.03 V

Calculate the standard electrode potential, $E^\ominus$, for the $\text{Cl}_2 / \text{Cl}^-$ electrode.

[1 mark]

$E^\ominus$ for the $\text{Cl}_2 / \text{Cl}^-$ electrode _____ V

11

Turn over ►



0	7
---	---

This question is about complexes with bidentate ligands.

0	7	.	1
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Explain what is meant by bidentate ligand.

[1 mark]

0	7	.	2
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When an excess of sodium ethanedioate solution is added to an aqueous solution of cobalt(II) chloride, a pair of stereoisomers is formed.

Draw the displayed formula of each isomer.
Show clearly the charge on each complex.

[3 marks]



0	7	.	3
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When an excess of ethane-1,2-diamine ($\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$) is added to a solution containing aqueous cobalt(II) ions, all the water ligands are replaced.

Write an equation for the reaction.

Explain the thermodynamic reasons why this reaction occurs.

[6 marks]

Equation

Explanation

10

Turn over for the next question

Turn over ►



0 8

This question is about elements in Period 3.

0 8 . 1

Write equations, including state symbols, to show the reaction of phosphorus with excess oxygen and the reaction of phosphorus with excess chlorine.

State **one** observation that you would make in **both** reactions.**[3 marks]**

Equation with oxygen

Equation with chlorine

Observation

0 8 . 2

Write an equation to show the reaction of aluminium oxide with hydrochloric acid.

[1 mark]

0 8 . 3

Aluminium oxide is added to an excess of water.

Describe what is observed.

[1 mark]



0	8	.	4
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Sulfuric acid can be used to test for barium compounds.
Phosphoric acid can be used in a similar way to test for barium compounds.

Write an equation for the reaction between barium hydroxide and phosphoric acid.

Suggest what you would observe in this reaction.

[2 marks]

Equation

Observation

7

Turn over for the next question

Turn over ►



0	9
---	---

This question is about the colours of transition metal ions.

0	9	.	1
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Explain why an aqueous solution of copper(II) ions is blue.

[2 marks]

$[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ has a single d-orbital electron.

The energy needed to excite this electron is $3.79 \times 10^{-19} \text{ J}$

0	9	.	2
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Calculate the wavelength, in nm, of light that will be absorbed by this $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ complex.

Planck's constant, $h = 6.63 \times 10^{-34} \text{ J s}$

The speed of light, $c = 3.00 \times 10^8 \text{ m s}^{-1}$

[2 marks]

Wavelength _____ nm



0 9 . 3

Aqueous ammonia is added a drop at a time to an aqueous solution containing copper(II) ions until the ammonia is in excess.

Describe what is observed.

Write equations for any reactions that occur.

[4 marks]

8

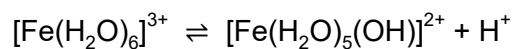
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1 0

This question is about the acidity of metal-aqua ions.

1 0 . 1Iron(III) compounds dissolve in water to form $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ions.The $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ion dissociates as shown.Write an expression for the acid dissociation constant, K_a , for the $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ion.Calculate the pH of a solution made when 1.35 g of iron(III) chloride are dissolved in water to make 750 cm^3 of aqueous solution.Give your answer to **two** decimal places.The value of $\text{p}K_a$ for the $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ion is 2.22**[6 marks]**Expression for K_a

Calculation

pH _____



1	0	.	2
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Solutions containing $\text{Fe}^{3+}(\text{aq})$, $\text{Fe}^{2+}(\text{aq})$ or $\text{Al}^{3+}(\text{aq})$ ions are acidic.

State why solutions of $\text{Fe}^{2+}(\text{aq})$ are **less** acidic than $\text{Fe}^{3+}(\text{aq})$

Suggest why solutions of $\text{Al}^{3+}(\text{aq})$ are **more** acidic than $\text{Fe}^{3+}(\text{aq})$

[2 marks]

$\text{Fe}^{2+}(\text{aq})$ is less acidic than $\text{Fe}^{3+}(\text{aq})$ _____

$\text{Al}^{3+}(\text{aq})$ is more acidic than $\text{Fe}^{3+}(\text{aq})$ _____

8

END OF QUESTIONS



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ANSWER IN THE SPACES PROVIDED**



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